Trapped-Ion Qudit Quantum Computing: Due-Diligence Assessment of the Ringbauer Program

Rowan Brad Quni-Gudzinas

2026-08-10

Abstract

This note documents a due-diligence assessment of the trapped-ion qudit quantum computing program led by Martin Ringbauer at the University of Innsbruck. Three publications anchor the assessment: the universal qudit processor demonstration (2022), the native qudit entanglement gate based on a generalized light-shift mechanism (2023), and a lattice gauge theory simulation with ion qudits (2024). All bibliographic metadata, author lists, DOIs, and experimental figures cited here were verified against live Crossref records and the published article body. The note records the measured light-shift gate fidelities by dimension, the quadratic scaling of gate error with qudit dimension, and the relevance of this empirical curve to the dimensional-advantage crossover parameter in qudit benchmarking.

1 Purpose and Scope

This note records a structured due-diligence assessment of the trapped-ion qudit quantum computing research program centered at the University of Innsbruck and associated with Martin Ringbauer. The assessment was conducted against the following primary sources, each verified live during the assessment session:

Source	Identifier	Verification
Universal qudit processor (2022)	DOI `10.1038/s41567-022-01658-0`	Crossref + arXiv API
Native qudit entanglement gate (2023)	DOI `10.1038/s41467-023-37375-2`	Crossref + full article body
Lattice gauge theory simulation (2024)	DOI `10.1103/PRXQuantum.5.040309`	Crossref (title + author list)

The scope is limited to (a) verifying the bibliographic record, (b) extracting the experimental gate-performance data, and (c) assessing the relevance of that data to qudit benchmarking questions. It is not a review of the full qudit literature.

2 Bibliographic Record

2.1 Universal Qudit Processor (2022)

- **Title:** A universal qudit quantum processor with trapped ions
- **Authors:** Martin Ringbauer, Michael Meth, Lukas Postler, Roman Stricker, Rainer Blatt, Philipp Schindler, Thomas Monz
- **Venue:** Nature Physics, volume 18, article 1053 (2022)
- **DOI:** `10.1038/s41567-022-01658-0`
- **arXiv:** `2109.06903` (submitted 2021-09-14)

The work demonstrates a universal qudit quantum processor with a local Hilbert space dimension of up to 7, using calcium-40 ions in a surface Paul trap. The authors report performance comparable to qubit-based processors while enabling native simulation of high-dimensional quantum systems and more efficient implementation of qubit-based algorithms. [established — abstract, arXiv API]

2.2 Native Qudit Entanglement Gate (2023)

- **Title:** Native qudit entanglement in a trapped ion quantum processor
- **Authors:** Pavel Hrmo, Benjamin Wilhelm, Lukas Gerster, Martin W. van Mourik, Marcus Huber, Rainer Blatt, Philipp Schindler, Thomas Monz, Martin Ringbauer (Hrmo and Wilhelm contributed equally)
- **Venue:** Nature Communications, volume 14, article 2242 (2023)
- **DOI:** `10.1038/s41467-023-37375-2`
- **Funding:** ERC Starting Grant QUDITS (101039522); ERC Consolidator Grant Cocoquest (101043705); EU H2020 MSCA 840450; EU Quantum Flagship AQTION

820495. and MILLENION; FWF SFB BeyondC (F7109); FWF START (Y879-N27); FFG 872766; US Army Research Office W911NF-21-1-0007 and W911NF-16-1-0070; ODNI/IARPA. [established — acknowledgements section of the article body]

2.3 Lattice Gauge Theory Simulation (2024)

- **Title:** Digital Quantum Simulation of a (1+1)D SU(2) Lattice Gauge Theory with Ion Qudits
- **Authors:** Giuseppe Calajó, Giuseppe Magnifico, Claire Edmunds, Martin Ringbauer, Simone Montangero, Pietro Silvi
- **Venue:** PRX Quantum, volume 5, article 040309 (2024)
- **DOI:** `10.1103/PRXQuantum.5.040309`

The 2024 work is a digital quantum simulation of a (1+1)-dimensional SU(2) lattice gauge theory using ion qudits. It is an original simulation study, not a review article. This classification is recorded explicitly because the work is occasionally miscited as a review of the qudit field. [established — Crossref title and author-list verification]

3 Experimental Gate Data

3.1 Encoding and Gate Mechanism

The qudit encoding uses the calcium-40 ion: the $|0\rangle$ state is the $S_{1/2}, m_j = -1/2$ ground level, and the states $|i\rangle$ with $i \in \{1, 2, 3, 4\}$ are Zeeman sub-levels of the $D_{5/2}$ manifold. Coherent operations between sub-levels use 729 nm laser light; 401 nm light generates the light shift for the state-dependent force. [established — article body]

The entangling gate is a generalized light-shift (LS) gate. The gate action is symmetric on all excited qudit states, which implies the same gate mechanism can be used irrespective of qudit dimension, and that the calibration overhead does not increase with dimension. The composite gate is constructed from d applications of the light-shift pulse $U_{\mathrm{LS}}(t_g)$ interleaved with cyclic permutation operators X_d:

$$G = (X_d U_{\mathrm{LS}}(t_g))^d$$

Up to a global phase, the gate acts as:

$$G(\theta) : |jj\rangle \to |jj\rangle, \qquad |jk\rangle \to e^{i\theta}|jk\rangle \text{ if } j \neq k$$

This action directly generates genuine qudit entanglement, as opposed to embedding qubit-level entanglement in a larger Hilbert space. [established — article body]

3.2 Measured Fidelities

The gate fidelity was extracted from exponential decay fits over repeated gate applications (up to 9 applications between state preparation and reversed preparation). The measured state fidelities for dimensions $d = 2$ through $d = 5$ are:

Dimension d	Measured fidelity
2	$99.6(1)\%$
3	$98.7(2)\%$
4	$97.0(3)\%$
5	$93.7(3)\%$

[established — article body, Figure 4 and main text]

The measured gate performance degrades quadratically with dimension. The paper's numeric error model reproduces the data with good agreement and attributes the dominant errors at higher dimension to Rabi frequency noise and slow frequency noise causing dephasing of local operations; at $d = 2$ the fidelity is limited by motional coherence time and gate-laser frequency noise. [established — article body]

3.3 Entanglement Properties

The entanglement of the states generated by a single gate application was evaluated using the Schmidt number, concurrence, and entanglement of formation. For all dimensions tested, the Schmidt number is maximal (equal to d), which certifies genuine qudit entanglement up to $d = 5$. For $d > 2$, the concurrence significantly exceeds the maximal possible value for any qubit state. [established — article body]

4 Relevance to Qudit Benchmarking

The dimensional-advantage crossover parameter d^* is the minimum qudit dimension at which the encoding-density benefit of a qudit (which carries $\log_2 d$ bits per physical system) overcomes the per-gate fidelity penalty; this framing follows the benchmarking framework defined in [1]. The measured fidelity staircase above provides an empirical anchor for this parameter: the per-gate error grows approximately quadratically in d while the encoding density grows only logarithmically. At the demonstrated operating point ($d = 3$ with 98.7% gate fidelity), the encoding-density gain is $\log_2 3 \approx 1.58$ bits per system, which offsets the reduced per-gate fidelity relative to qubits only if the algorithmic depth is correspondingly reduced or the encoding overhead is otherwise exploited. The 2024 lattice gauge theory simulation is a concrete demonstration of the reduced circuit-depth argument: qudit encoding directly compresses the simulation register for a non-abelian gauge theory.

No quantitative claim about a specific crossover value is made in this note; the purpose of Section 4 is to record the empirical fidelity curve as a benchmark input. [my conjecture — the framing of d^* as a benchmark input]

5 Verification Statement

All bibliographic metadata (titles, author lists, DOIs, venues, volumes) and all experimental figures (fidelities, Schmidt number, gate mechanism, funding acknowledgements) were verified during the assessment session against at least one live source: Crossref API, arXiv API, or the full published article body. No citation in this note is drawn from unverified memory of the literature.

6 Limitations

- The assessment covers only the three primary sources listed in Section 1.
- The PRX Quantum (2024) record was verified via Crossref metadata (title and author list) but the article body was not read in full during this assessment.
- The fidelity extraction methodology is summarized from the article; the numeric error model is cited but not independently recomputed here.
- This note does not review competing qudit platforms (photonic, superconducting, neutral-atom) or the broader qudit error-correction literature.

References

1. Quni-Gudzinas, R. B. The Qudit Advantage: System-Level Joules-per-Solution Comparison of a Qudit Architecture Against 17 Conventional Qubit Quantum Computing Platforms. Zenodo, doi:10.5281/zenodo.21827737 (2026).